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CONSTRUCTION
SERVICES

GRAYCO STUDY GUIDE

Fire & Smoke Restoration Key Terms

Every key term from the course with a plain-English definition, grouped by subject. Use it to review before the GrayCo test and the IICRC FSRT exam.

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KEY TERMS

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SUBJECTS

GRAYCO CONSTRUCTION SERVICES, LLC · BARTLETT, TENNESSEE

STUDY GUIDE

Key terms and definitions

01 Introduction

Pretesting
Testing a material or surface for how it responds to cleaning and deodorizing. Only after identifying and pretesting does the restorer decide what gets restored and what gets replaced.

Insured
The property owner, the primary customer.

Insurance carrier
The insurance carrier's representative is the secondary customer when the owner is insured. The whole process should be agreed with all materially interested parties.

02 Fire and Smoke

Incomplete combustion
Smoke and soot particles are the visual by-products of incomplete combustion.

Combustion
The self-sustaining process of rapid oxidation. Needs three elements: fuel, oxygen and heat.

0.1 to 4 microns
Size range of smoke particles (particulate matter) produced by fire. One micron = 1/25,000 of an inch; invisible to the naked eye.

Soot particles / PAHs
Smoke residues fall into two groups: PAHs (polycyclic aromatic hydrocarbons, a VOC) and soot particles.

Carcinogenic
Both soot particles and PAHs have been reported to be carcinogenic, so everyone on a fire site must wear properly fitted respirators with HEPA-rated filters.

Heightened awareness odor
Also called psychological odor. People who were in the structure during the fire stay sensitive to the odor even after it is gone. The most difficult odor for a contractor to overcome.

Protein residues
From overcooked or burned meat, fish or beans. Yellow to brown, greasy, sticks to most surfaces. Removed with thorough water-based cleaning if the surface is wet-cleanable.

Natural substance odor and residue
From burned paper or wood (campfires, forest fires, fireplaces). Grey to black, dry and powdery. Vacuum / dry precondition, then wet clean. Easiest of the three to clean.

Synthetic residues
From burned oil-based materials (carpet, upholstery, drapes, furniture, toys, plastics). Thick, heavy, black, smeary. Do not touch; vacuum it away or it smears.

Air pressures and temperatures
The factors that determine how far smoke penetrates through a structure.

Severity of soot contamination
Directly related to combustion time, heat and presence of oxygen. Less oxygen = more soot; plenty of oxygen = less soot.

Natural wicks
Porous materials that hold onto smoke odor: fabrics, unfinished wood, paper, unsealed concrete.

Oily film
Vaporized oils carried in smoke settle as an oily film that makes residue hard to clean.

Yellow
Light-colored plastics exposed to high heat and vaporized oils can stain or turn yellow; this cannot economically be reversed.

03 Safety Preparations

Everyone
Safety on the job site is the responsibility of everyone.

Equipment and safety training
Federal agencies (OSHA in the U.S.) require firms to issue safety equipment and provide safety training.

SDS (safety data sheets), labels, training
The three ways to get information about chemical hazards: SDSs from the manufacturer, product labels, and training from the manufacturer or supplier.

Labeled
All restoration chemicals must be kept in properly labeled containers and secured.

In or anywhere around
Chemically sensitive customers should not be in or anywhere around affected areas until restoration is complete. Get informed consent before using deodorants.

Chemically sensitive
A customer who reports symptoms from very low chemical exposure. May need special air scrubbing, cleaning agents and deodorizers.

HEPA air scrubbers
Air quality control to use when working around loose soot, along with exhaust fans.

Personal protective equipment (PPE)
The last line of defense after engineering controls. Level of PPE is based on the initial hazard inspection and risk assessment.

Asbestos and lead-based paints
Two common invisible safety threats in older structures.

Lead
Heavy metal used in paint made before 1978.

Asbestos
Insulating and fire-retarding material used in siding, roofing, flooring and adhesives before 1977.

Demolition
Lead and asbestos are most likely released during sanding and demolition. If asbestos is suspected, stop and have it tested by a qualified lab.

HEPA filtration
The most important PPE around toxic particles is a respirator with HEPA filtration.

04 Emergency Response

Attitude
The most important asset a technician can bring to the job site is a proper attitude.

Documentation
Photograph and document all pre-existing conditions. Customers who witness documentation are less likely to complain about pre-existing damage or lost items. It protects you.

Insured or property owner
The customer with financial responsibility who signs the work authorization (a legal binding agreement). No work begins until it is signed.

Structural integrity
What to watch for, along with other safety hazards, when entering a fire-damaged structure.

Cold surfaces
Hot smoky air is drawn to cold surfaces: outside walls, glass, metal. Also check horizontal surfaces and plastics that carry static.

Durable protection / minimal additional damage
What a board-up must provide. Methods: cover-over, fitted inserts, bolt tension, tarps, shrink-wrap.

Loss mitigation services
Winterization to prevent freezing, corrosion control, and cleaning surfaces to prevent staining.

Acidic
Soot combined with atmospheric water vapor becomes acidic.

Corrosion
Neutralizing acid smoke residue (usually with alkaline solutions) is the first stage of corrosion prevention.

05 Initial Mitigation Services

Elevated
Odors are most apparent when humidity and temperature are elevated, as right after a fire.

Odors and additional soiling
Removing charred, unsalvageable materials early helps eliminate odors and additional soiling.

Combination
Intense smoke odor needs a combination of techniques and procedures.

Pairing and masking agents
What most deodorants contain. Pairing agent joins with the odor to reduce or eliminate it. Masking agent covers it with a stronger, pleasant scent.

Ultra low volume (ULV) fogger
Venturi atomizer making 8–15 micron droplets that can stay airborne for hours; applies water-based deodorants and encapsulants.

Thermal fogger
Uses heat to make a dry, very fine (about 0.5 micron) fog that behaves like smoke; white, warm, attracted to cool surfaces.

Oxidizers
Chlorine bleach (liquid), ozone and hydroxyl radicals (gases).

Chlorine bleach
Liquid oxidizer (sodium hypochlorite). Discolors nylon, dissolves protein fibers like wool and silk, corrodes some metals, dangerous with ammonia.

Ozone (O3)
A gaseous oxidizing agent, produced on site by an ozone machine (electrical arc splits O2). Can harm people, pets, live plants and natural rubber. Works slowly, 24–72 hours.

Hydroxyl / ultraviolet
Hydroxyl generators use ultraviolet light plus water vapor to make hydroxyl molecules that oxidize odors.

Pairing agent
A product that combines with a malodor to form a new odorless compound.

Water based odor counteractants
Mixed with water for pressure spraying or added to water-based cleaning solutions on surfaces water won't hurt.

Solvent based thermal fog solutions
Used in a thermal fogger or added to solvent-based cleaning solutions.

Heavy duty solvent odor counteractants
Also called dry solvent. Contact application on some wood surfaces or added to solvent cleaning solutions and sealers.

Measure / label instructions
To ensure proper concentration, measure properly and follow manufacturer label instructions. Ask the customer before using additives.

Four principles of odor control
1) Remove the source (if possible). 2) Clean the source area. 3) Penetration. 4) Sealing surfaces.

Heightened awareness (early deodorization)
Deodorizing early reduces heightened awareness odor by lowering odor intensity.

Begin
Odor removal should begin in the room or area where the fire started.

Masking agents (granules, oils, gels)
Fragrance-impregnated oils, gels, solids or granules that prolong evaporation of deodorant vapors.

Thermal fogging
Before thermal fogging, contact and advise the appropriate agencies (fire, police, security).

Possible explosion, respiratory irritation, slippery residues
Hazards of thermal fogging petroleum-based deodorizers excessively in confined areas.

Vacated
Areas treated with ULV fog, thermal fog or ozone must be vacated by anyone not authorized or not wearing proper PPE.

Properly documented
Food in boxes or cans exposed to heat is properly documented before disposal.

Lamb's wool duster / vacuuming
Precondition all pack-out contents with a lamb's wool duster or by vacuuming with a brush attachment.

06 Principles and Methods for Cleaning

Washable
Surfaces not adversely affected by water-based (wet) cleaning.

Non-washable
Surfaces that cannot be cleaned with wet methods.

Preconditioning
Removes soot from delicate fabrics and surfaces by dry soot removal: lamb's wool duster, HEPA vacuum with soft brush, or compressed air.

With water, then chemical concentrates
When mixing, fill the container with water first, then add the concentrate, so you get the right mix without foam.

Wood restoring cream
For finished wood furniture and hard surfaces with heavy smoke damage; always work with the direction of the wood grain.

07 Structural Cleaning

Cleaning
Usually the most economical restoration option.

Fabrics and surfaces
The restorer's main responsibility is to preserve fabrics and surfaces from further damage.

Source
In light-to-moderate odor jobs, begin by cleaning the source area.

Heat line
A visible broad line where high temperatures were present. Its location affects how restorable surfaces are, and how furniture is judged.

Deteriorate
Acoustical (popcorn) ceilings will deteriorate when cleaned aggressively.

Secure breakables
Before wet cleaning, secure breakables and remove drapes, wall décor and framed items.

Ceiling to floor
Structure cleaning starts at the ceiling, moves to middle and lower surfaces, and finishes on the floor.

Bottom to top
The exception: painted walls are washed from bottom to top to reduce streaking.

General purpose cleaner or degreaser
Removes moderate soot from gloss or semi-gloss enamel paint. General purpose cleaners also clean high-gloss paint, plastic and porcelain.

Pre-condition (flat latex)
On flat latex paint with light-to-moderate soot, pre-condition or dry sponge first. Wet cleaning means repainting.

Sealer
If odors remain after cleaning painted surfaces, apply a sealer (a stain sealer covers stains and seals odors).

Attic complications
Stored contents of questionable value, rough surfaces, and hard to reach areas.

Encapsulation
A procedure that interrupts off-gassing of smoke odor; including one material inside another so it is not apparent.

Blast
Charred wood: abrasively blast or wire brush off the charring, or scrape and seal, or remove and replace.

Customer
The biggest mistake is not pretesting surfaces or not getting the agreement signed by the customer.

Wood grain
Wipe finished wood in the same direction as the wood grain to hide minor scratches.

Mild alkaline detergent
Wet clean finished paneling with a mild alkaline detergent or commercial panel cleaner, then polish.

Wood restoring cream, gel or compound
Use on built-in cabinetry and furniture where smoke is embedded in a clear finish.

Surface is dry
Inspect masonry only after it is dry; wet masonry looks clean.

Alkaline solutions and media blasting
The two common methods for cleaning masonry.

Acid based / etching
If alkaline fails, use acid-based cleaners. Acid removes a microscopic layer of masonry, called etching.

Sealer or finish
Masonry floors may be coated with a clear sealer or finish that needs extra cleaning.

Media blasting
Most aggressive method for severely stained brick and mortar. Media: sand, glass, dry ice, soda, sponge, nut shells, corn cob.

Vinyl wall coverings
Durable, made for water-based cleaning; use a moderate to highly alkaline product after vacuuming.

Paper wall coverings
Less common now; found more in older buildings.

Flocked wall covering
Velvet-like material glued to a paper backing for a raised texture.

Grass cloth
Natural woven covering; one of the hardest surfaces to treat; dyes damage easily.

End of the job
Salvageable finished floors are cleaned at the beginning and again at the end of the job.

Fiberglass
The bath fixture most easily stained by smoke. Clean with non-abrasive cream, foam or liquid soft-scrub.

Heavy duty alkaline or mild acid
Tile grout cleaners. Acid cleaners also work on ceramic tile and aluminum framing; try alkaline first.

Ventilate and don appropriate PPE
Required when using aggressive acid tile cleaners in enclosed areas.

Vacuuming
First step for non-asbestos acoustical ceiling tile with very light soot, then dry sponge.

Turn off (and tag out)
First step for light fixtures: turn off and tag out or secure the circuit.

Kitchen ventilation system
Clean vent hoods inside and out, exhaust pipes/ducting, and replace hood filters.

Fans and exhaust duct systems
Odor sources often overlooked, along with hood filters.

Mounting cavities and subsurface areas
Clean these when cleaning built-in appliances.

Puff-back
A malfunction of an oil-fired heating system.

Inspection
An HVAC inspection determines when the whole system needs cleaning.

NADCA standards
National Air Duct Cleaners Association. Heavily sooted HVAC components are disassembled, cleaned and deodorized to NADCA standards.

Duct types
Vinyl flex duct (insulation outside, covered in vinyl); rigid fiberglass duct board (foil outside); galvanized sheet metal (insulation inside).

08 Contents Cleaning

Monetary, antique and sentimental
The three types of value to weigh when deciding time, effort and cost to restore an item.

Foam cleaning
Gentle method; applies only foam, not liquid, to moisture-sensitive surfaces.

Spray and wipe
Limits liquid and dwell time: spray with a hand sprayer and wipe immediately.

Abrasive cleaning
Aggressive; for surfaces not hurt by agitation. Uses abrasive liquids/pastes with steel wool or pads.

Immersion cleaning
For items that can be fully submerged, like dishes and colorfast clothing (pre-soaked).

Document
Non-restorable contents must be documented before disposal, with customer approval.

Finished wood
Sealed and coated with a clear or opaque protective finish.

Unfinished wood
Bare and unsealed; adsorbs more smoke odor than finished wood.

0000 steel wool
The finest, least aggressive grade, used with cleaner on wood finishes.

Clear finish
Wood with a clear finish responds best to wood restoring gel, cream or compound.

Heat and moisture absorption
Cause of white discoloration on finished wood; treat with wood cream and 0000 steel wool.

Furniture polish
Final step to restore luster.

Wood finish (stacking)
Don't stack furniture right after wood restoring cream; it can distort the wood finish.

Refinishing
Blistered, charred or permanently stained wood goes to a professional refinishing contractor.

Light sanding
May remove small remaining smoke stains on unfinished wood.

Darken
Petroleum-based deodorant can darken porous wood; never use without customer approval.

Fiber and weave, preexisting conditions, degree and type of soot damage
Considerations before cleaning upholstery.

Dry soil removal
Normally the first step in restoring smoke-damaged fabric.

Dry solvent
Recommended where wet cleaning risks shrinkage, browning or bleeding.

Dry cleaning
Waterless cleaning liquids with hand or machine agitation.

Dry vacuuming, air washing, solvent cleaning
How newer, lightly or moderately sooted upholstery can be cleaned without browning, bleeding or shrinking.

Powder cleaning
Powder brushed in with a natural bristle brush, dried, then vacuumed.

Manual wet cleaning
Shampoo or low-moisture foam, with deodorant in the solution.

Pre-spray / brush agitate / hot water extraction
Most common method for fire-damaged upholstery.

As soon as practical
Severely smoked clothing is cleaned and deodorized as soon as practical; remaining garments after initial clothing needs, too.

Select
After inspecting soft goods, ask the customer to select what to save and what to discard.

Disposed of immediately
Metal wire and plastic hangers, because of low value.

Ticking
The cover fabric on a mattress; usually colorfast (color won't fade or run when wet cleaned).

Pre-existing stains
Stains on used mattresses that soot removal won't fix.

Uneven tops and bottoms, weak fabric, watermarks
Pre-existing drapery conditions to note before cleaning.

Protected surface or ultrasonic
Where blinds are wet cleaned and deodorized.

Colored, decorative trim
Causes problems when wet cleaning fabric lampshades. Handle shades by the top of the metal frame only.

Inspected first, then cleaned
Electronics: inspect, then clean and test by a qualified technician.

Inspected (toys)
Toys: inspect, wet clean, rinse and sanitize.

Oil and acrylic
Paintings that are more restorable than watercolor, charcoal or pastel.

Document (artwork)
Inspect high-value artwork and document its condition, preferably with photos.

Turned over
Personal papers removed from desks are turned over to the client.

Inspecting books
Check: type of construction materials, evidence of moisture, degree of heat and smoke damage.

Dry sponge, crumbly dough cleaner, art gum eraser
Dry cleaning tools for preconditioned books and page edges.

Working files / stored files
Working files are used daily and restored first; stored files are lower priority.

Dishes
Clean as soon as practical; presoak heavy soot in a degreaser; wrap in white newsprint or bubble wrap.

Potential damage (china and crystal)
Disclaim liability for damage not visible during inspection. Clean china and crystal by hand only, never in a dishwasher.

Contents categories
Scheduled contents (valuables), collectable, general, electrical and electronic, water-damaged contents.

Avoid surface-to-surface contact
When transporting hard furnishings back. Pad and strap each piece before loading.

Protect
In storage, keep aisles and protect contents from re-contamination.